

Bluffing with Precision: LLM-Guided Strategy and Opponent Modeling in Multi-Agent Poker

Yanny Gao¹ Sara Kothari¹

¹Department of Computer Science, Stanford



Introduction

In reinforcement learning, solving partially observable, imperfect-information, zero-sum games such as poker presents a core challenge, as agents must act under uncertainty and hidden opponent information. A particularly difficult aspect is enabling AI agents to exhibit strategic deception and follow specific playing styles in competitive multi-agent settings. While state-of-the-art poker AI systems can play at high levels, they often lack the capacity to adaptively deceive opponents using nuanced bluffing strategies, limiting their effectiveness in dynamic and adversarial scenarios.

This capability is critical for building AI that can operate effectively in real-world settings involving competition, negotiation, or security, where opponents may act unpredictably or deceptively. Enabling adaptive deception and strategy adherence improves robustness, realism, and generalization in multi-agent interactions. We present a novel approach that integrates four key innovations:

- **Curriculum Learning:** Adaptive opponent difficulty progression based on agent performance metrics
- **Online Opponent Modeling:** Transformer-based system predicting opponent actions, bluff tendencies, and hand strength during gameplay
- **Bluff-Aware Rewards:** Custom reward functions that incentivize successful deceptive play through Monte Carlo hand evaluation
- **Natural Language Feedback:** GPT-3.5 generated strategic critiques embedded via sentence transformers to guide policy improvements

Our system demonstrates emergent deceptive behaviors while maintaining strong performance against diverse opponent types, advancing the state-of-the-art in strategic AI for competitive environments.

Related work

While prior works like DeepStack[1] and AlphaHoldem [3]have achieved strong performance in heads-up no-limit Texas Hold’em using CFR and end-to-end RL, they remain computationally intensive and lack real-time adaptability. AMP3 [2] introduces opponent style modeling for multi-player poker, but relies on static, offline models and predefined behavior libraries. Recent benchmarks like PokerBench [4]show that even state-of-the-art LLMs struggle with strategic poker play, underscoring a gap in adaptive reasoning. Our approach fills this gap by performing online opponent modeling, deception-aware learning, and incorporating natural language feedback from an LLM—e.g., “Too loose preflop, adjust aggression post-flop”—to iteratively refine strategy/style during training and enable richer behavioral/policy adaptation.

Experiment

- We use Petting Zoo’s Texas Hold’em No Limit environment, a variation of Texas Hold’em where there is no limit on the amount of each raise or the number of raises. This is a 2 player game.
- **Online Opponent Modeling:** Each agent maintains a transformer-based OpponentModel with OpponentTracker that learns to predict opponent’s next actions (and hand strength, bluff tendency, aggression) incorporating real-time action prediction into agent decision-making with separate models trained for different opponent types.
- **Comprehensive Statistics Tracking:** Real-time computation of professional poker metrics with stage-aware action tracking across preflop, flop, turn, and river betting rounds. Performance analytics measured across different opponent types.
- **Bluff-Aware Reward System:** We implement deception-based rewards through compute_deception_reward() function when agents take aggressive actions with low hand strength. Hand strength is estimated via Monte Carlo simulation with 20 rollouts using estimate_bluff_score(). System provides both intermediate rewards for aggressive low-strength actions and delayed rewards when successful bluffs lead to episode wins.
- **Three-Tier Curriculum Learning:** Adaptive opponent progression system with WeakOpponent (random policy), MediumOpponent (pre-trained actor-critic agent, no updates), and StrongOpponent (learns alongside main agent via REINFORCE updates). Promotion thresholds set at 70% win rate (weak→medium) and 65% win rate (medium→strong) based on empirical performance.
- **Natural Language Feedback Integration:** Policy critiques generated by GPT-3.5 after each episode based on professional poker statistics (VPIP, PFR, AFq, WTSD) are encoded via sentence transformers into embeddings and integrated into agent’s observation space for continuous policy improvement and possible dynamic policy/style switching.
- **Architecture & Hyperparameters:** Shared two-layer MLP policy network. Learning rate = 0.01, discount factor $\gamma = 0.99$, bluff reward weight = 1.0, bluff win bonus multiplier = 0.2.

Results and Discussions

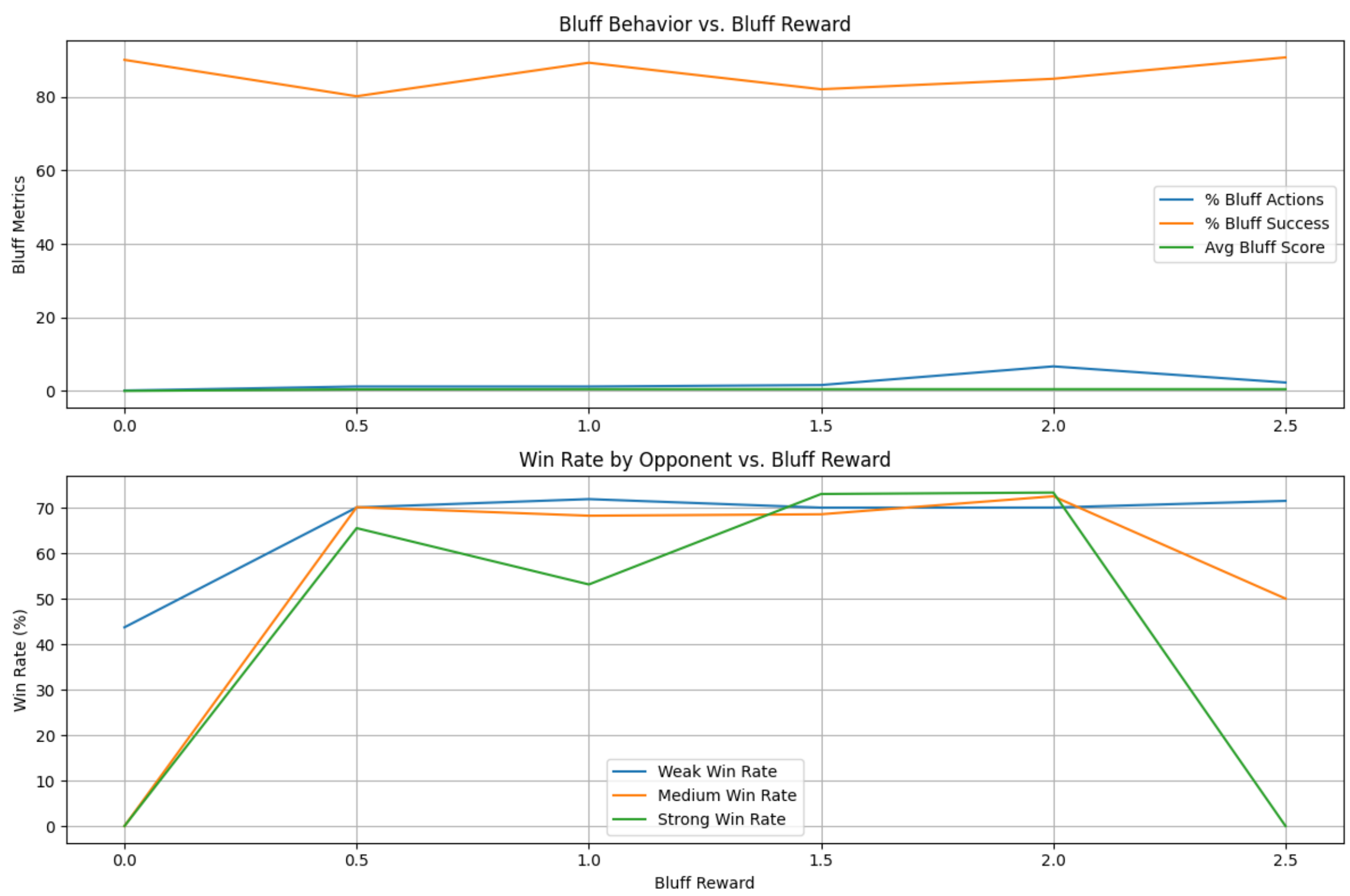


Figure 1. Strategic progression of bluff-related metrics during training

Bluff rewards improve both bluff frequency and success. A reward of 2.0 achieves the best balance, with 6.68% bluff rate, 84.95% success, and a 73.32% win rate against strong opponents. Without rewards, bluffing vanishes and win rates suffer.

Model	% Bluff	% Success	Avg Score	Weak Win	Medium Win	Strong Win
With Opponent Modeling	8.21	77.49	0.475	71.88%	68.23%	53.12%
Without Opponent Modeling	9.30	52.30	0.468	70.00%	71.00%	33.90%
Δ	-11.72%	+48.16%	+1.50%	+2.69%	-3.90%	+56.69%

Table 1. Performance Comparison With and Without Opponent Modeling

Opponent modeling improves bluff quality and boosts strong-opponent win rate to 72.76% (+56.69%). Despite fewer bluffs, success rate climbs to 84.69%, with an average bluff score of 0.474. By Episode 9000, the model achieves 0.2218 opponent accuracy, enabling sharper decisions under pressure.

Tier	% Bluff Actions	% Bluff Success	Win Rate (%)
Weak	Static: 2.21%	76.47%	71.88
	Curriculum: 1.20%	89.32%	70.00
	Δ : -45.70%	+16.78%	-2.61%
Medium	Static: 2.22%	77.93%	68.23
	Curriculum: 1.20%	89.32%	70.35
	Δ : -45.95%	+14.58%	+3.11%
Strong	Static: 8.21%	77.49%	53.12
	Curriculum: 1.20%	89.32%	72.76
	Δ : -85.38%	+15.27%	+36.94%

Table 2. Bluff Rate, Success, and Win Rate: Curriculum vs Static Training

Curriculum learning consistently improves bluffing performance across all difficulty tiers. Across the board, bluff success rates increase significantly while the overall frequency of bluff attempts is reduced. It enables the agent to build robust tactics early on and adapt them effectively in more complex scenarios, resulting in smarter, not riskier, bluffing decisions under pressure.

Opponent	Baseline	With GPT	Change (Δ)
WeakOpponent	74.50%	71.88%	-2.62%
StrongOpponent	53.12%	73.77%	+20.65%

Table 3. Win Rate Comparison vs. Weak and Strong Opponents (Before vs. After GPT Feedback)

We observe that LLM feedback consistently encourages tighter, more controlled play, with common critiques like "Agent needs to tighten pre-flop range, bluff less, control aggression," resulting in measurably decreased AFq (Aggression Frequency). This strategic guidance demonstrates tangible impact: win rates against the strong opponent improved dramatically from 53.12% to 73.77% when combining bluff rewards (weight=1.0) with opponent modeling and LLM feedback integration, confirming that natural language guidance effectively shapes both agent strategy and performance outcomes.

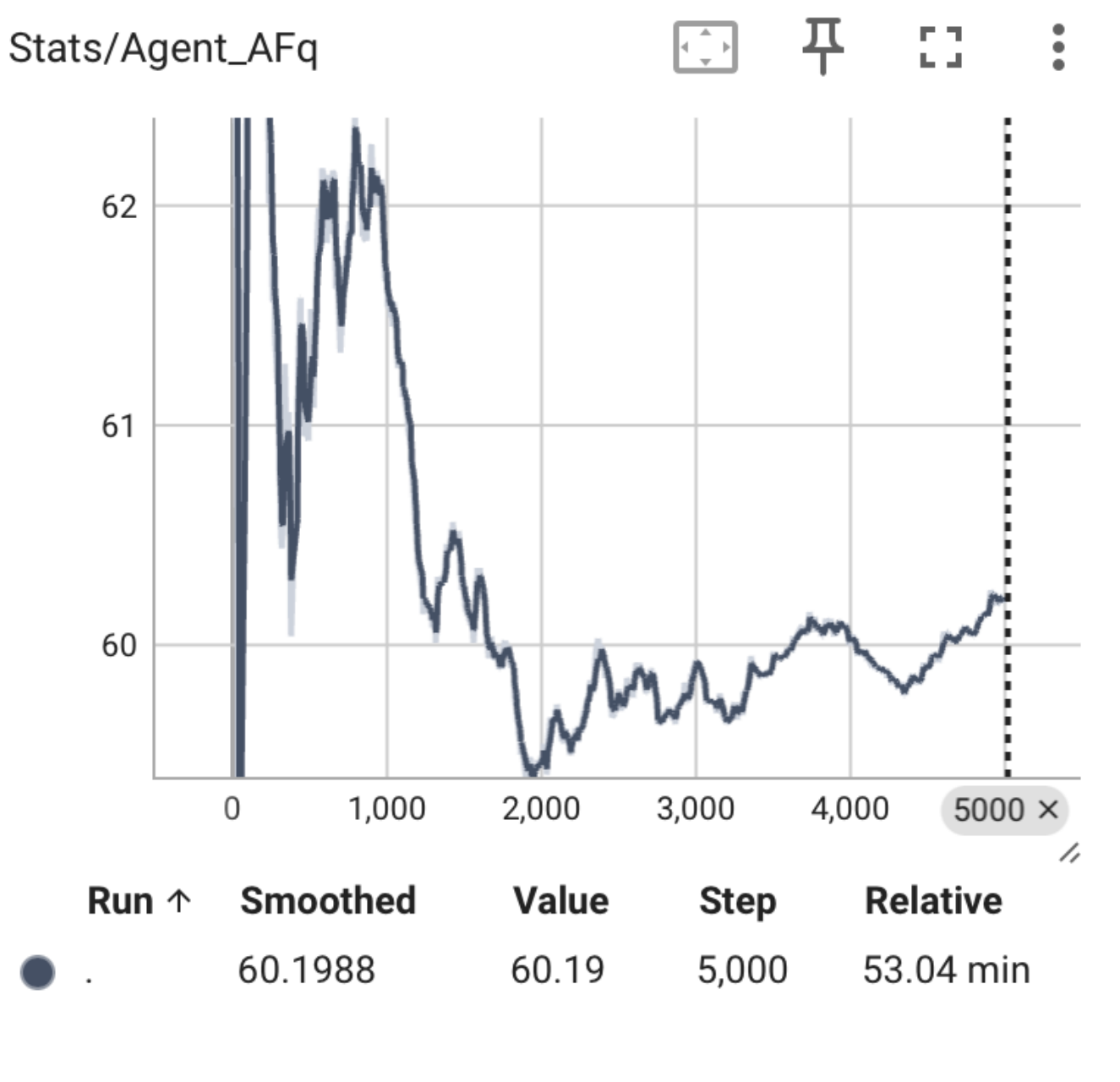


Figure 2. Aggression Frequency (AFq) over training episodes with LLM feedback integration

Limitation and Future

We plan to enhance our transformer-based opponent modeler for better prediction accuracy and implement contrastive loss between natural language feedback embeddings and agent statistics to ensure behavior alignment. While our current system uses implicit policy switching through LLM feedback, we propose explicit encouragement via additional rewards/losses that directly measure and incentivize adherence to LLM strategic advice, combined with auxiliary rewards based on real-time poker metrics (VPIP, PFR, AFq, WTSD) for more targeted behavioral modification/ adaptive style change.

References

- [1] DeepStack.
- [2] Daming Shi, Xudong Guo, Yi Liu, and Wenhui Fan. Adaptive multi-player poker policy learning based on opponent style modeling.
- [3] Enmin Zhao, Renye Yan, Jinqiu Li, Kai Li, and Junliang Xing. AlphaHoldem: High-performance artificial intelligence for heads-up no-limit poker via end-to-end reinforcement learning. 36(4):4689–4697. Number: 4.
- [4] Richard Zhuang, Akshat Gupta, Richard Yang, Aniket Rahane, Zhengyu Li, and Gopala Anumanchipalli. PokerBench: Training large language models to become professional poker players.

1. **VPIP:** percentage of hands a player decides to voluntarily fund. $VPIP = \frac{\text{Hands played pre-flop}}{\text{Total hands dealt}} \times 100$
2. **PFR:** pre-flop aggressiveness of a player. percentage of all the actions that are aggressive $PFR = \frac{\text{Hands raised pre-flop}}{\text{Total hands dealt}} \times 100$
3. **AFq:** percentage of all the actions that are aggressive. $AFq = \frac{\text{Bluffs}}{\text{Total Hands Dealt}} \times 100$
4. **WTSD:** percentage of a player playing to showdown $WTSD = \frac{\text{Hands went to Showdown}}{\text{Total hands went to Flop}} \times 100$